Class 10: Drude Model: Source of shortcomings

To understand the source of shortcomings in the Drude model, let us numerically estimate some of the important quantities predicted by the Drude model. These quantities are as follows:

$$\sigma = \frac{N_f e^2 \tau}{m}$$

$$K = \frac{1}{3}\langle v^2 \rangle \tau C_V^e$$

$$C_V^e = \frac{3}{2} N_f k_b$$

$$\frac{1}{2} m \langle v^2 \rangle = \frac{3}{2} k_b T$$

Consider the metal silver, which is a good conductor of electricity and heat, and let us carry out calculations with respect to this element.

Element: Ag
Atomic weight: 108 amu
Density: 10.5 gm/cc

The density is therefore:

10.5×10^6 gm/m^3

The number of moles per cubic meter is given by:

$10.5 \times 10^6/108 \sim 10^5$ moles/m^3

The corresponding number of atoms per cubic meter is given by:

$10^5 \times 6.023 \times 10^{23}$ atoms of Ag/ m^3

$= 6.023 \times 10^{28}$ atoms of Ag/ m^3

Assuming that the Ag atoms are univalent, the number of free electrons per cubic meter will be the same as the number of Ag atoms per cubic meter

$$\therefore N_f = 6.023 \times 10^{28}$$

Since $k_b = 1.38 \times 10^{-23}$ J/atom/K

$$C_V^e = \frac{3}{2} N_f k_b = \frac{3}{2} \times 6.023 \times 10^{28} \times 1.38 \times 10^{-23}$$

$$\therefore C_V^e \approx 12.5 \times 10^5 \text{ J/m}^3\text{/K}$$

Since there are approximately 10^5 moles/m^3 of free electrons in solid Ag (same as the number of moles of Ag atoms per cubic meter),

$$C_V^e \approx 12.5 \text{ J/mole/K}$$

Data obtained at low temperatures in the 1960s indicates that the experimental value for C_V^e is ≈ 0.2 J/mole/K

Therefore the theory overestimates the value of C_V^e by approximately two orders of magnitude.

Since the estimation of thermal conductivity is correct, and is given by the equation

$$K = \frac{1}{3} \langle v^2 \rangle \tau C_V^e$$

It appears that we are underestimating $\langle v^2 \rangle$ by two orders of magnitude – assuming that we are estimating τ correctly, which we will see shortly.

The value of $\langle v^2 \rangle$ predicted by the Drude model, may be calculated as follows:

$$\frac{1}{2} m \langle v^2 \rangle = \frac{3}{2} k_b T$$

Given that the mass of the electron $= 9.1 \times 10^{-31}$ kg, at room temperature, or approximately 300 K, we have

$$\langle v^2 \rangle = \frac{3 \times 1.38 \times 10^{-23} \times 300}{9.1 \times 10^{-31}} \text{ m}^2/\text{s}^2$$

Therefore, as predicted by the Drude model,

$$\langle v^2 \rangle \approx 1.38 \times 10^{10} \text{ m}^2/\text{s}^2$$

Which we find is an underestimate by two orders of magnitude.

Let us now estimate drift velocity of electrons in Ag that has been subject to a potential difference across its ends. We will need to make some assumptions about the dimensions of the wire and the potential difference being applied across it. Let us assume that we have a silver wire that has a length $l = 1\ m$ and a diameter of 1 mm (10^{-3} m), and that we have applied 1 V across the length of the wire.

Given that the conductivity, σ, of Ag is $= 6.3 \times 10^7\ \Omega^{-1}\mathrm{m}^{-1}$, we have the resistance R of the wire given by

$$R = \frac{\rho l}{A} = \frac{l}{\sigma A}$$

Where ρ is the resistivity, and A is the cross sectional area of the wire.

$$\therefore R = \frac{1}{6.3 \times 10^7 \times \pi \times 10^{-6}/4} \approx \frac{1}{50}\ \Omega$$

When we apply 1 V across the ends of this wire, using Ohm's law we have:

$$V = IR$$

$$\therefore I = 50\ amperes$$

Which, incidentally, is a lot of current, but is to be expected since Ag is good conductor of electricity.

Assuming that all of the free electrons are carrying the current, and assuming that there is conservation of charge, i.e., the charge entering the wire is equal to the charge leaving the 1 m length of wire, then, in the time taken for electrons to travel the length of the wire, the total charge entering or leaving the length of wire is equal to the number of free electrons in the length of wire.

The number of free electrons in the wire $= N_f \times l \times A$

Charge corresponding to these electrons $= N_f \times l \times A \times e$

In the presence of the voltage, if we assume that the electrons have a drift velocity v_f, time taken by the electron to travel a distance l is l/v_f. If the current in the circuit is I, the total charge transported during this time is $I \times l/v_f$

Therefore:

$$I \times \frac{l}{v_f} = N_f \times l \times A \times e$$

Rearranging and simplifying, we get:

$$v_f = \frac{I}{N_f \times A \times e} \approx \frac{50}{6 \times 10^{28} \times 0.8 \times 10^{-6} \times 1.6 \times 10^{-19}}$$

Therefore

$$v_f = 6 \times 10^{-3} \text{ m/s}$$

Or, the drift velocity is approximately 6 mm/s

It is quite interesting to discover that value of the drift velocity is so low. At this velocity, it will take approximately 150 seconds, or two and a half minutes, for the current to travel the 1 m length of the wire. This drift velocity is much slower than normal human walking speed! Yet our day to day experience is contrary to this. As soon as we turn on the switch, the light bulb comes on, no matter how far it is from the switch. How is this possible given the low drift velocity of the electrons? The answer is simple, electrons close to the bulb start moving as well and light the bulb, electrons from the switch get there much later. The situation is the same as water in a pipe that is already full of water – just the way a wire is full of free electrons from end to end. When the tap is opened far away, water that is already close to the exit of the pipe begins to flow out of the pipe almost immediately. Water just exiting from the tap takes much longer to reach the end of the same pipe.

While discussing the topic of electron velocity, we must note the difference between the drift velocity we have just calculated and the mean square velocity we calculated a little bit earlier. Drift velocity is the net velocity of the electrons in the direction of the applied field and it turned out to be a relatively small number. The mean square velocity, corresponds to the velocity with which electrons randomly move within the material – there is no net movement of electrons in any direction in this case. The mean square velocity turned out to be a large number, which itself we indicated had been under estimated by two orders of magnitude.

The drift velocity is a believable number since it is extracted directly from the current flowing in a conductor, assuming simply charge conservation. The mean time between collisions, τ, can be directly estimated from v_f, as we will see below. Therefore we can place confidence in the estimate of τ, and hence we are justified in doubting the estimate of $\langle v^2 \rangle$, as we indicated earlier.

Estimating τ:

$$\tau = \frac{m v_f}{\varepsilon e} = \frac{9.1 \times 10^{-31} \times 6 \times 10^{-3}}{1.6 \times 10^{-19} \times 1} = 3.5 \times 10^{-14} \text{ s}$$

The relationship used to determine τ, as shown above, comes from the experiments associated with movement of electrons in response to an electric field. However, we are 'placing confidence' on its use in relation to the expression for thermal conductivity, where there is no electric field applied. Why is this so?

The justification for this can be thought of as follows: Electrons are moving very rapidly, and in a random manner as a result of their thermal energy. This results in a high value for $\langle v^2 \rangle$, which is of the order of $10^{10} \text{ m}^2/\text{s}^2$, which can be loosely thought of as

corresponding to a modulus of velocity of the order 10^5 m/s. Due to electrons moving around with such high velocities within an enclosed region, *due to their thermal energy*, they collide with each other and this results in the mean time between collisions that is being identified as τ. The additional velocity that the applied electric field causes is relatively miniscule, and is of the order of 10^{-3} m/s, in the direction of the applied field. This drift velocity $v_f$in response to the electric field, is limited by the collisions that the electrons encounter as a result of the random movement of the electrons due to their thermal energy. In other words, the collisions are dominated by the thermal state of the material. Therefore, even though we use the drift velocity (caused by the applied electric field) to gauge the meantime between collisions, the value obtained is dominated by the thermal state of the system, and therefore can be used for describing the thermal behavior with confidence. In other words, it is the v_f of the electrons that is limited by the existence of a τ in the system, rather than the τ being limited by the v_f of the electrons.

In summary, through our discussions above, we find that the Drude model over estimates the specific heat at constant volume of the free electrons, and underestimates the mean square velocity, or effectively the translational kinetic energy of the electrons. In other words the Drude model is not able to accurately describe how energy is present within the system.

For the Drude model, the Kinetic theory of gases has been used to describe how energy is held within the system of free electrons in the solid conductor. Given the very large number of free electrons in a conductor, and the large number of atoms in a mole of an ideal gas, the kinetic theory does not try to determine the velocities of each electron or atom individually. Instead a statistical description is used to understand how energy is held within the system. This statistical description, used for ideal gases, is called the Maxwell-Boltzmann distribution. It applies to a large collection of non interacting particles, such as the atoms of an ideal gas, and estimates how many particles will occupy an energy level, or posses a certain amount of energy. Or in other words how particles are distributed across energy levels. From this the $\langle v^2 \rangle$ can be calculated.

To the extent that we have imposed the kinetic theory of gases on electrons in a solid, we have effectively imposed the Maxwell-Boltzmann statistics on the collection of free electrons.

The errors we see in the Drude model indicate that the assumption of Maxwell-Boltzmann statistics for the free electrons, is incorrect at some fundamental level. This is the underlying message we have extracted from our analysis of the system so far. This is not surprising because we have imposed rules that pertain to non interacting particles, on electrons which could interact with each other or with the ionic cores.

To arrive at improved models for electrons in a solid, in the next class we will look at the assumptions of the Maxwell-Boltzmann distribution and derive the distribution. We will examine the results we will obtain. We will recognize that only certain types of particles can be reasonably expected to follow the Maxwell-Boltzmann statistics. These are called Classical particles. We will focus our attention on the definition of a classical particle and

consider what other description will be reasonable for electrons. Once we arrive at an alternate description for electrons, we will again try and make predictions and examine how successful we have been in improving the model for electrons in a solid.